

Daniil Katulevskiy

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AI/ML & Performance-focused Software Engineer – building high-throughput, GPU-accelerated systems, distributed Kubernetes infrastructure, and computer vision/LLM pipelines; targeting roles at the intersection of machine learning, systems performance, and platform engineering.

EDUCATION

University of California, San Diego - B.S., Computer Science Jun 2026

- GPA: 3.91. Top 5% of class. Coursework: Deep Learning, Distributed Systems, Blockchain Security, Large Language Models, Networked Systems, GPU Programming, Embedded systems. *M.S. Machine Learning expected Jan 2027*

EXPERIENCE

UCSD x Google Phone-Based Datacenter - Researcher Mar 2026 - Present

- Contributing to a UCSD-Google project building a ~20,000-phone datacenter on Google Pixel hardware for AI inference and serverless compute workloads.
- Building a minimal Debian image for the Pixel SoC, designed to boot stateless over PXE so the fleet can run Kubernetes with Karmada federation across the 5,000+ devices per cluster.
- Working at the bootloader/kernel layer: brought up UART debug access on device, ported Android-specific kernel modules into mainline Linux kernel, and am building toward unattended network-boot provisioning.

Polity Network - Platform Engineer Jun 2024 - Present

- Wrote architecture documentation for a 6-microservice fintech platform (Kotlin/Spring Boot, Go, Python), covering service boundaries, data flow, auth model, and the EC2-to-Kubernetes migration plan.
- Migrated the platform from a single-EC2 monolith to AWS EKS; built OpenTofu modules for the cluster and ArgoCD pipelines with ephemeral per-PR review environments, cutting release cycle time by ~40%.
- Implemented hybrid RBAC/ABAC authorization with centralized policy enforcement, integrated Infitical with the Kubernetes External Secrets Operator, remediated 60+ tracked security findings across services.

Triton Robotics, UC San Diego - AI & Vision Engineer Sep 2023 - Jun 2024

- Led a 6-engineer CV team. Finetuned YOLO-v8 on custom robotics imagery to ~25% higher mAP over the baseline, then quantized to FP16/INT8 via TensorRT for <30 ms inference on NVIDIA Jetson AGX.
- Wrote custom CUDA preprocessing kernels to reduce CPU-GPU transfer overhead, roughly doubling end-to-end inference throughput on the Jetson.

Skolkovo Institute of Science and Technology - AI/ML Research Intern Jun 2023 - Aug 2023

- Developed a camera-only navigation stack for autonomous traversal of unmapped, GPS-denied terrain; compared A*, D* Lite, and POMDP planners in real-time simulation, reaching up to 95% obstacle evasion.

PROJECTS

Custom Game Engine & Multiplayer 3D FPS - Entire game stack from scratch Mar 2026 - Present

- Built a cross-platform multiplayer 3D shooter and the underlying C++ engine from scratch; runs on Linux, Windows, and macOS via SDL3/SDL_GPU, which dispatches to Vulkan, DirectX 12, and Metal natively.
- Authoritative-server netcode with client-side prediction, server reconciliation, and lag compensation; PBR rendering, cascaded shadow maps, GTAO, SMAA T2x, SSR, volumetric lighting, and tonemapping.

GPU Kernel Optimization - CUDA / OpenCL BLAS Nov 2024 - Mar 2026

- Optimized dense GEMM kernels using shared-memory tiling, warp-shuffle reductions, and register micro-tiling; Nsight profiling and roofline analysis on RTX 3090 and 4090, reaching up to 85% of cuBLAS SGEMM throughput.

Multi-Region GPU Inference Cluster - Self-Hosted Kubernetes for LLM Workloads Jun 2024 - Present

- Built and operate a self-hosted Kubernetes cluster spanning 6 GPUs across multiple sites, used for LLM inference with vLLM and LoRA finetuning experiments on custom datasets; Prometheus/Grafana tracking.

Secure Messaging Platform - Post-Quantum Encrypted Chat May 2023 - Aug 2025

- Engineered E2E encrypted messaging with Kyber-1024 KEM, X3DH + Double Ratchet, identifier-free relays, and custom TLS obfuscation (anti-DPI). Go backend, TypeScript/React/Electron frontend.

TECHNICAL SKILLS

Programming Languages: C++, CUDA, Python, Go, Rust, TypeScript, Java, Solidity, x86/ARM ASM

GPU / HPC: CUDA kernels, OpenCL, cuBLAS/cuDNN, TensorRT, Nsight Systems, NCCL, roofline modeling

AI / ML: PyTorch, vLLM, ONNX Runtime, LoRA/QLoRA, FAISS, YOLO, sensor fusion

Infra: Kubernetes (bare-metal + EKS), OpenTofu/Terraform, ArgoCD, Helm, AWS, GitHub Actions CI/CD

Security: RBAC/ABAC, threat modeling, post-quantum crypto, reverse engineering, kernel debugging